Definitions

Digital

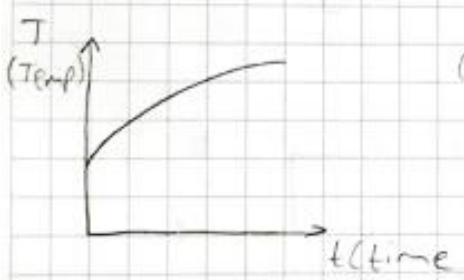
Logic

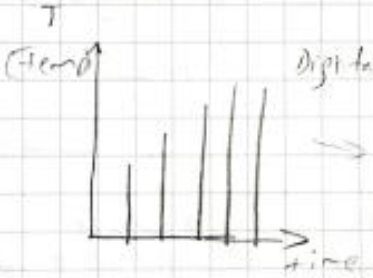
Gate

Digital and Analog quantities

Logic Levels

- Activation level
- Positive and Negative Logic





Digital Signal

→ Discrete quant

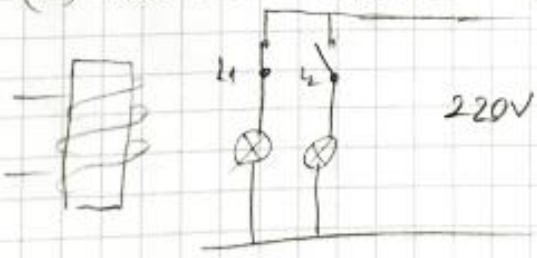
Zaman bakımından sürekli olan devrelere analog devreler denir.

Örneklenmiş, ayrılmış, quantalanmış ise ⇒ digital devreler

Logic Levels

(H) High = 1 = Set = True = High Voltage (5V) 4.0, 4.4 > 4

(L) Low = 0 = Reset = False = Low Voltage (0V) 4.5, 4.6, 5.0 > 5

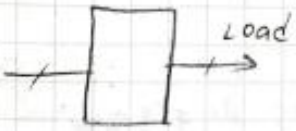


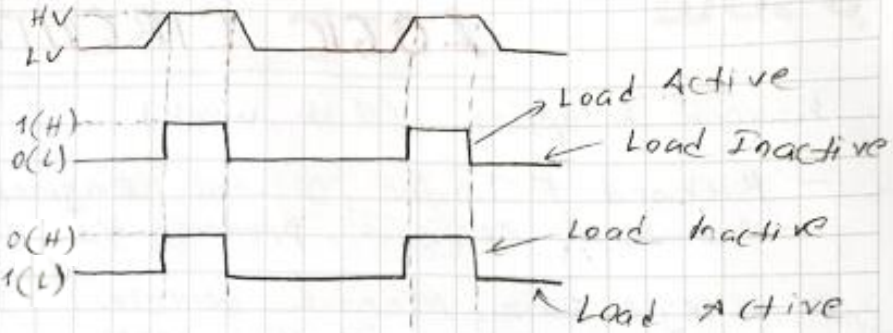
$V_{control} = 0$ L_1: on L_2: off

$V_{control} = 5V$ L_1 = off L_2 = on

L_1 (L) $V_c = 0$

L_2 (H) $V_c = 5V$





Logic Levels
- HV ⟷ Logic 1
- LV ⟷ Logic 0

} Positive Logic

- HV ⟷ Logic 0
- LV ⟷ Logic 1

} Negative Logic

Positive logic

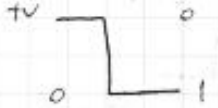


Positive potential (Source)



Negative potential (Source

Negative Logic



Positive potential (Source



Negative potential (Source

example

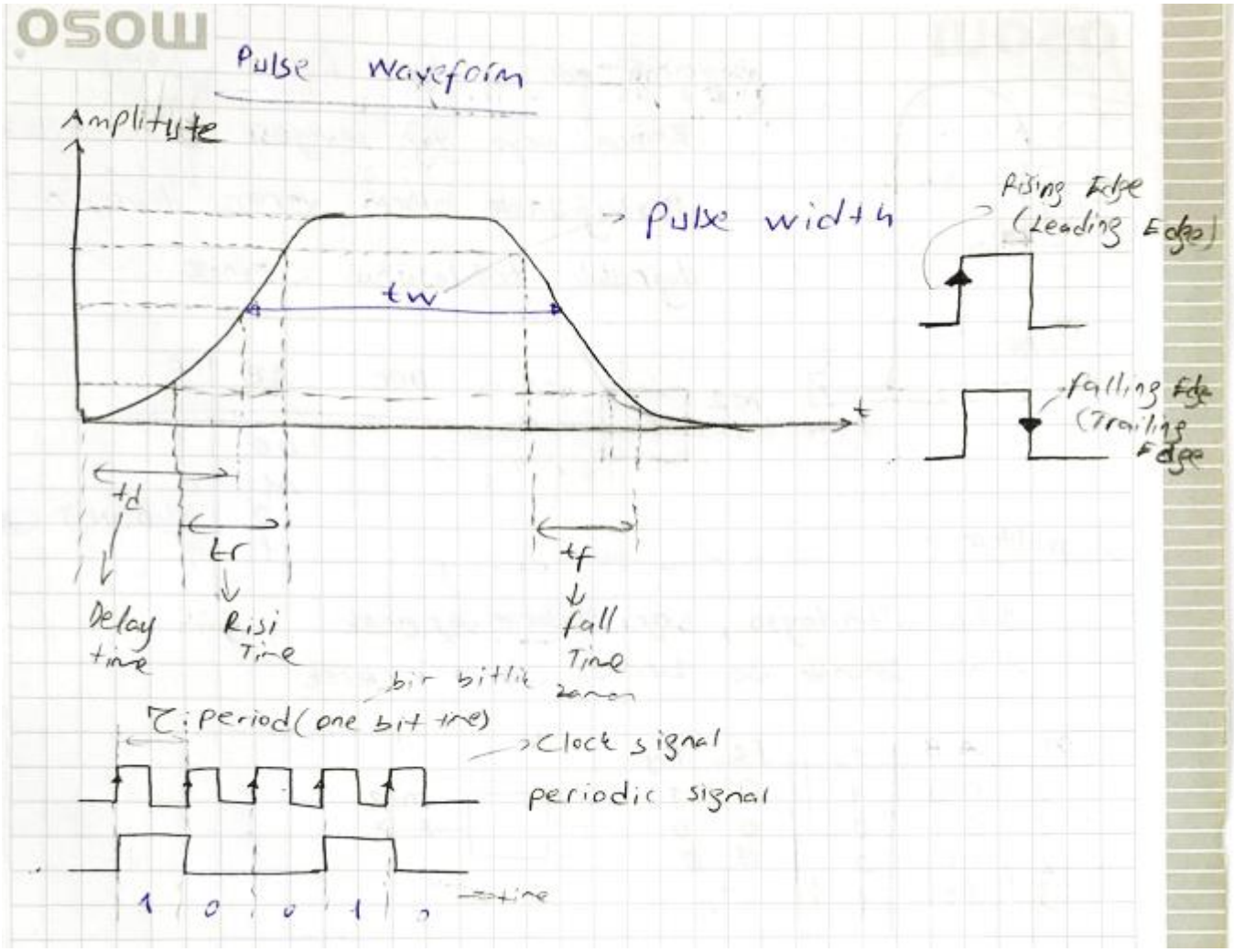
HLLH

PL → 1001

NL → 0110



	TTL	CMOS
	5V	5 V
	2,0	3,5
	0,8	1,5
	0	0


Pulse Waveform
Amplitude
→ Pulse width
tw
t
td
tr
tf
Delay time
Rise Time
fall Time
Rising Edge (Leading Edge)
falling Edge (Trailing Edge)
T: period (one bit time)
Clock signal
periodic signal
1 0 0 1 0
time